

Lecture 13 – Hydropower

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ECE180J

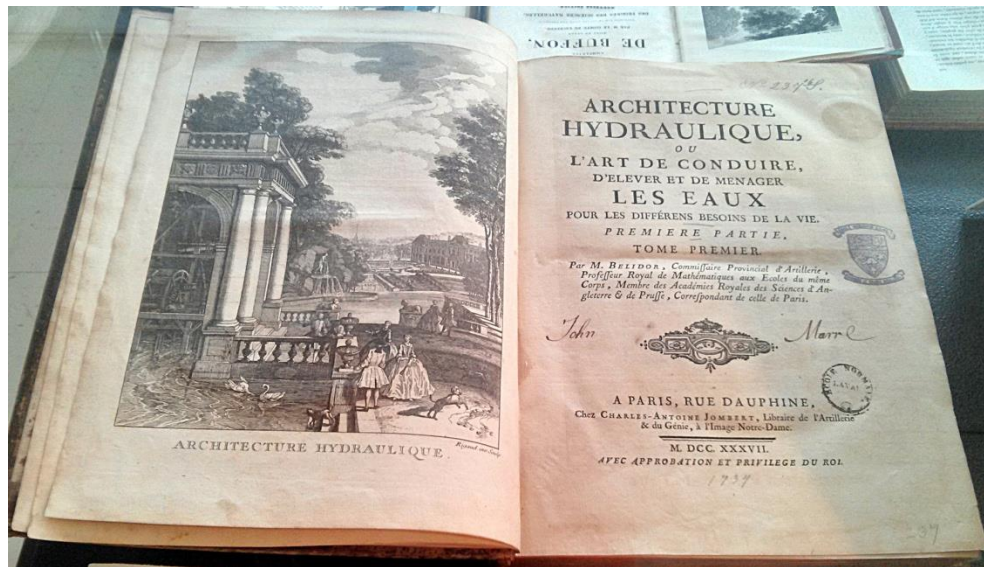


Outline

- Hydropower History and Resources
 - Types of Hydro Plants
 - Potential Energy, Flow Rate and Net Head
 - Types of Hydropower Turbines
 - Hydropower Economics
 - Environmental and Social Impacts
-
- B3: Chap 6
 - B1: Sec 8.5-8.6
 - REN21: Renewables 2025 Global Status Report

Long History of Water Power

- Hydropower is the oldest large-scale renewable electricity technology.
- Water wheels were used over 2,000 years ago, and modern hydropower turbines emerged in the mid-1700s.
- 1893: First U.S. commercial installation of AC hydropower plant at the Redlands Power Plant in CA.



Architecture Hydraulique written
by Bernard Forest de Bélidor



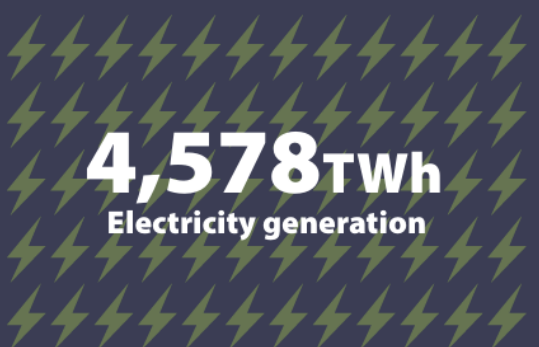
Waterwheel

Hydropower's global impact in numbers (2024)



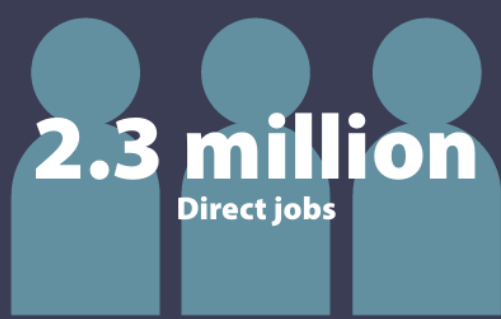
1,443GW

Installed hydropower capacity
(conventional + PSH)



4,578TWh

Electricity generation



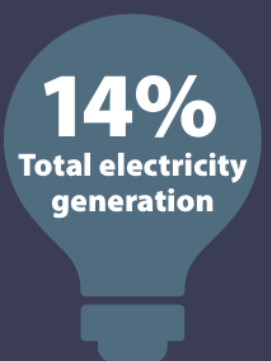
2.3 million

Direct jobs



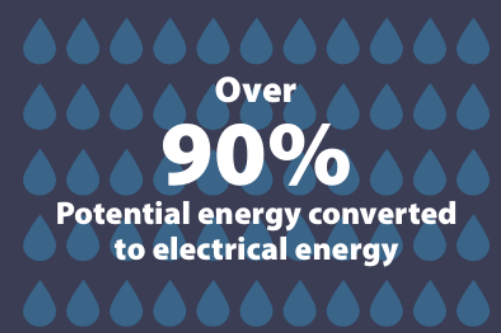
1,078GW

Capacity in
development



14%

Total electricity
generation



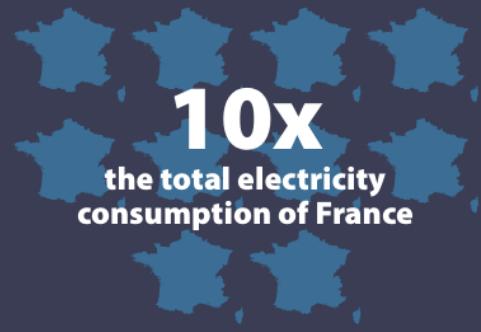
Over
90%

Potential energy converted
to electrical energy



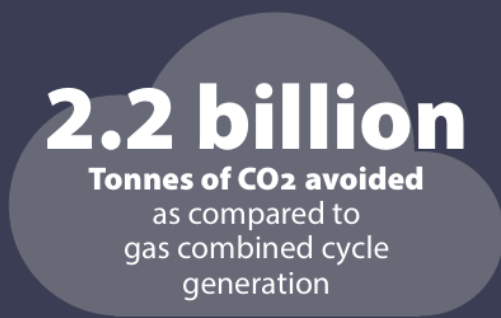
**100+
years**

Facility lifetime



10x

the total electricity
consumption of France



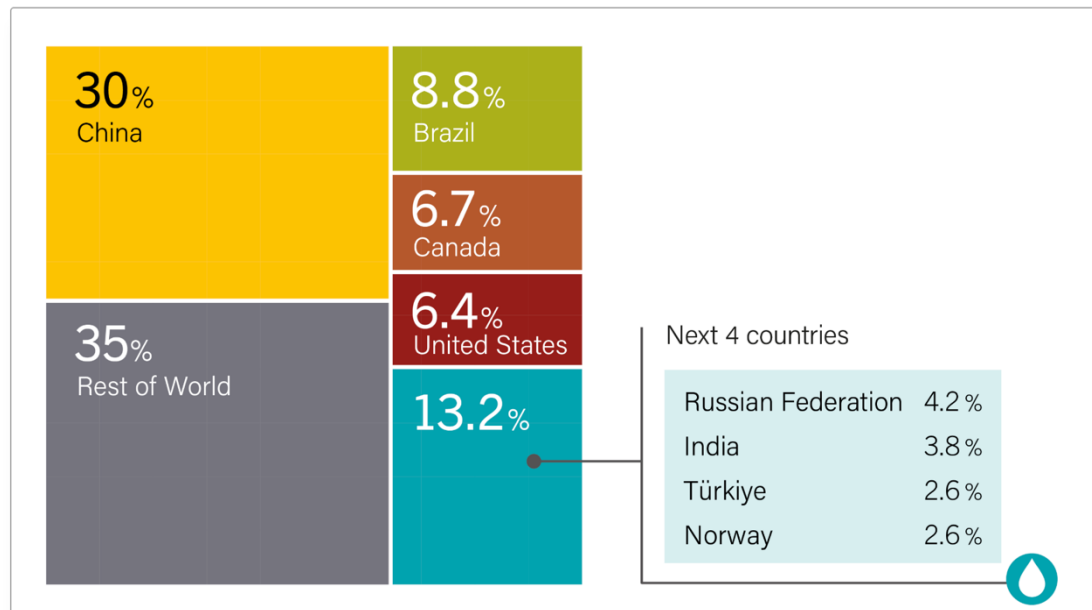
2.2 billion

Tonnes of CO₂ avoided
as compared to
gas combined cycle
generation

Hydropower Stats

FIGURE H-1 Global Hydropower Capacity, Shares of Top 8 Countries and Rest of World, 2024

Hydropower make up
47%
of all renewable
power generation.

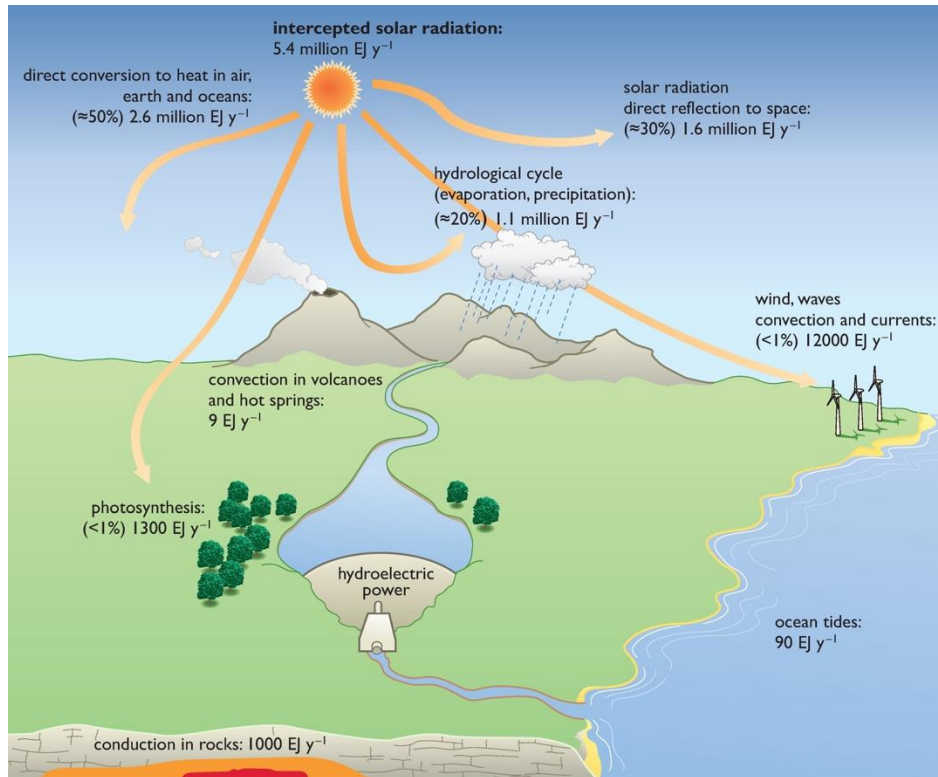


90 % of capacity
additions were in Asia
and Africa

103 GW of hydropower
capacity was added in the
last 5 years (2019-2024);
half of which in China.

11 countries each
installed at least 300 MW
of new capacity in 2024.

Hydropower Potential



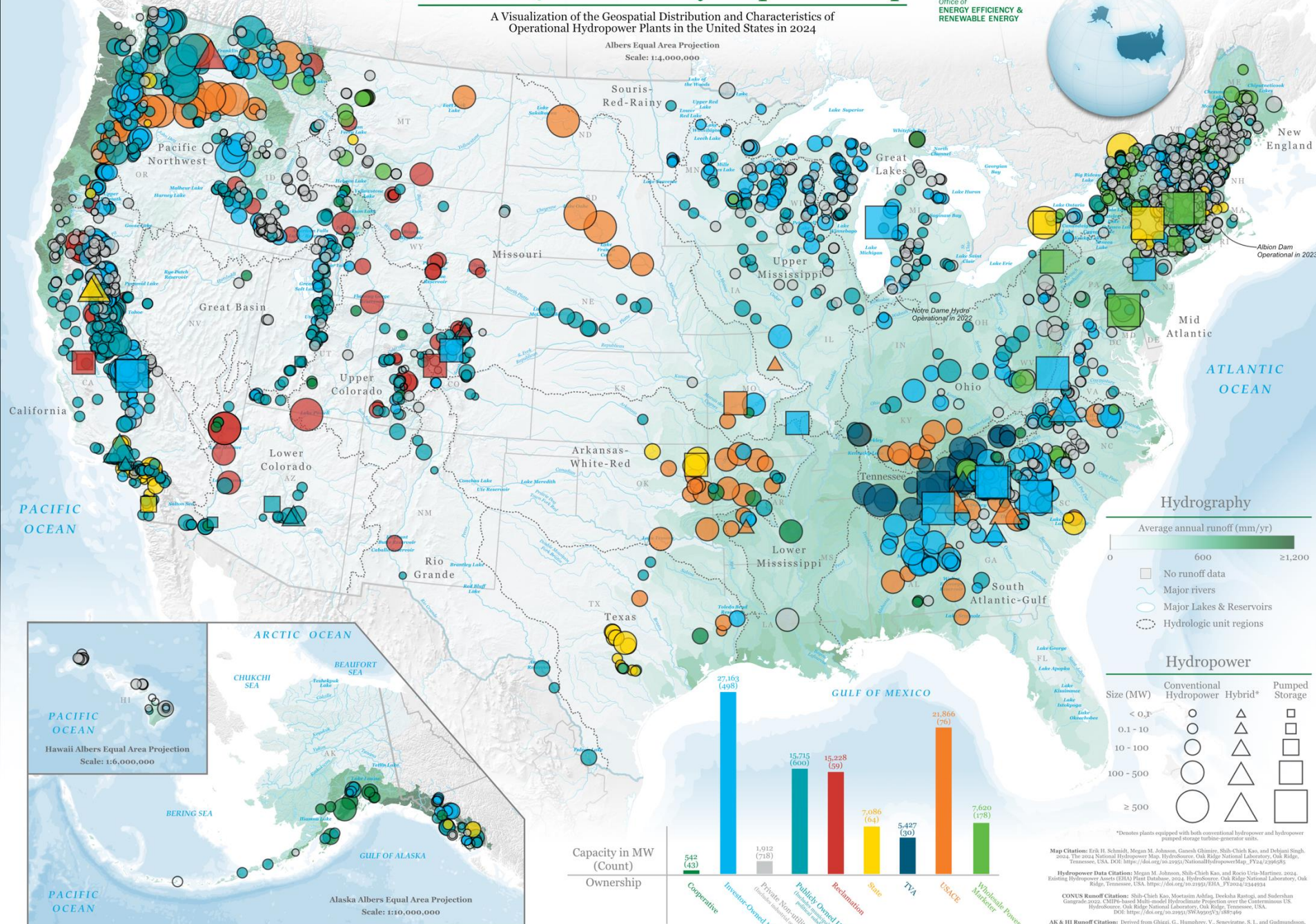
- About 25% of the solar energy reaching Earth drives water evaporation.
- Most of this energy is lost as heat when water vapor condenses. Only a tiny fraction (~200,000 TWh/year) returns to Earth as rain or snow and becomes potential hydropower.

- 20% (40,000 TWh/y) forms streams/ivers → **total hydro resource**.
- Impossible to build hydro plants everywhere → 40% (16,000 TWh/y) is regarded as the world's **technical hydro potential**.
- **Economic potential**: how much is available at a cost that is competitive with other energy sources.

The 2024 National Hydropower Map

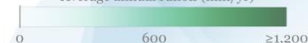
A Visualization of the Geospatial Distribution and Characteristics of Operational Hydropower Plants in the United States in 2024

Albers Equal Area Projection
Scale: 1:14,000,000



Hydrography

Average annual runoff (mm/yr)



- No runoff data
- Major rivers
- Major Lakes & Reservoirs
- Hydrologic unit regions

Hydropower

Size (MW)	Conventional Hydropower	Hybrid*	Pumped Storage
< 0.1	○	△	□
0.1 - 10	○	△	□
10 - 100	○	△	□
100 - 500	○	△	□
≥ 500	○	△	□

*Denotes plants equipped with both conventional hydropower and hydropower pumped storage turbine-generator units.

Capacity in MW
(Count)



Map Citation: Erik H. Schmidt, Megan M. Johnson, Ganesh Ghimire, Shih-Chieh Kao, and Dehshini Singh. 2024. The 2024 National Hydropower Map. HydroResource, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA. DOI: <https://doi.org/10.20932/HydroResource.2024.0001>

Hydropower Data Citation: Megan M. Johnson, Shih-Chieh Kao, and Reico Uri-Martin. 2024. Existing Hydropower Assets (EHA) Plant Database, 2024. HydroResource, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA. DOI: <https://doi.org/10.20932/HydroResource.2024.0002>

CORUS Runoff Citation: Shih-Chieh Kao, Mostafae Ashraf, Divesha Rastogi, and Sudeshan Gangrade. 2022. CMIP6-based Multi-model Hydroclimate Projection over the Continental US. HydroResource, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA. DOI: <https://doi.org/10.20932/HydroResource.2022.0001>

AK & HI Runoff Citation: Derived from Gage, C., Humphrey, V., Souvaine, S. L., and Gudmundsson, L. 2023. An observation-based global gridded runoff dataset from 1902 to 2014. Earth Syst. Sci. Data, 11, 1653-1674. DOI: <https://doi.org/10.5194/essd-11-1653-2019>

Small-Scale Hydro (SSH)

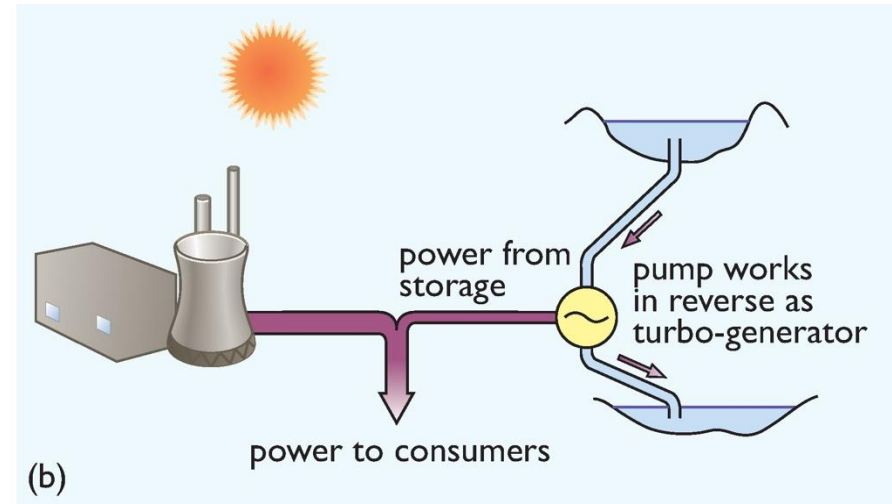
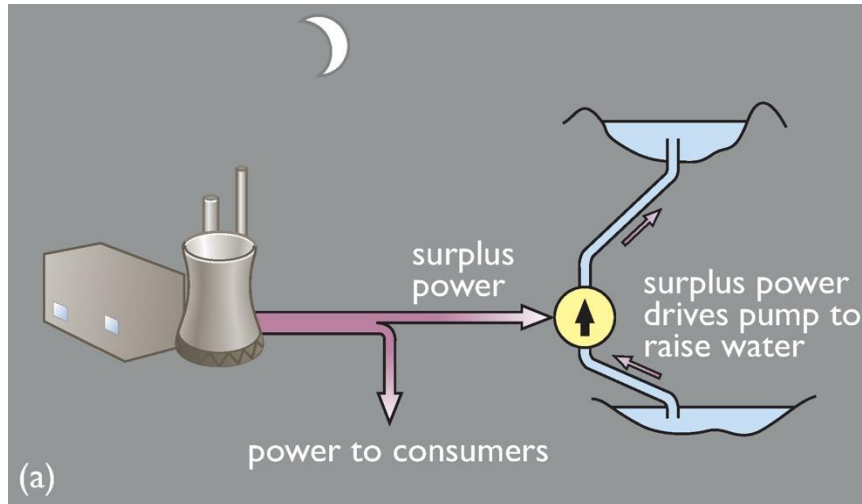
- Typically refers to hydropower plants with installed capacity < 10 MW
- Some countries use higher thresholds (e.g., 25–50 MW)

Latest Statistics:

- Global SSH capacity: ~79 GW as of 2022
- China accounts for over 50% of global SSH capacity
- SSH is widely used for rural electrification and distributed renewable energy

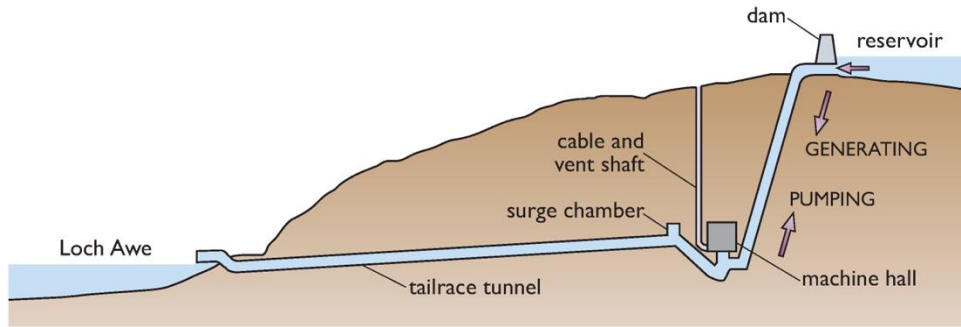


Pumped Storage Hydropower (PSH)



- The turbine can also run in either direction, either extracting energy from the water as a turbine or delivering energy to the water as a pump.
- PSH reached 182 GW globally by the end of 2023:
 - 1) ~13% of the total 1,412 GW hydropower capacity
 - 2) accounts for 76% of global utility-scale storage

PSH Example



(a)



(b)

Fig: Cruachan plant in Scotland:
440MW capacity, 705 GWh annual output

- Turbines/generators: very efficient (80%)
- Value of the system is enhanced by its speed of response: useful as back-up in case of a failure in the grid.
- Low-level reservoir of at least the capacity of the upper one must be available/constructed.
- PSH provides approximately 90–95% of the world's installed long-duration energy storage capacity

Huge Dams!



Hoover Dam: In the Black Canyon of the Colorado River

Commission date	1936–1960
Hydraulic head	590 ft (180 m) (Max)
Turbines	13 × 130 MW 2 × 127 MW 1 × 68.5 MW 1 × 61.5 MW Francis-type 2 × 2.4 MW Pelton-type
Installed capacity	2,078.8 MW
Capacity factor	23%
Annual generation	3.3 TWh (12 PJ) (2020) ^[3]

The Three Gorges Dam: in Yichang, Hubei province, China.

Commission date	2003–2012
Type	Conventional
Hydraulic head	Rated: 80.6 m (264 ft) Maximum: 113 m (371 ft) ^[3]
Turbines	32 × 700 MW 2 × 50 MW Francis-type
Installed capacity	22,500 MW
Capacity factor	45%
Annual generation	101.6 TWh (366 PJ) (2018)

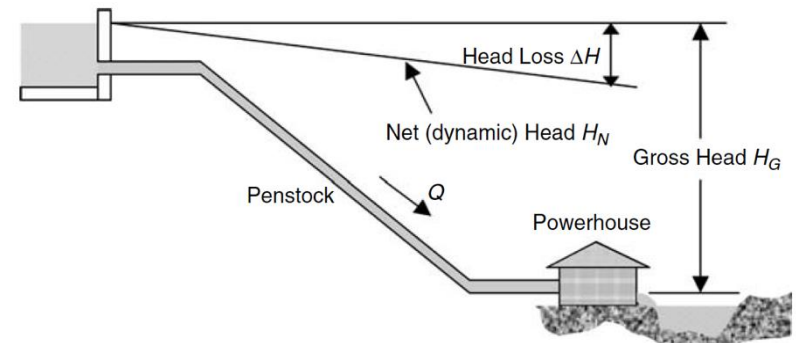


Head

Classify hydroelectric installations by:

- **effective head of water**
- **type of turbine used**
- capacity – the rated power output
- location and type of dam, reservoir, etc.

Head: the change in water levels (vertical height) between the hydro intake and the hydro discharge point.



$$\text{net head} = \text{gross head} - \text{friction head}$$

- gross head (actual elevation difference).
- friction head (head loss in piping): a function of the pipe diameter, the flow rate, the pipe length, smoothness of the pipe (material), how many bends, valves, and elbow the water has to pass through.

Barrage vs Dam

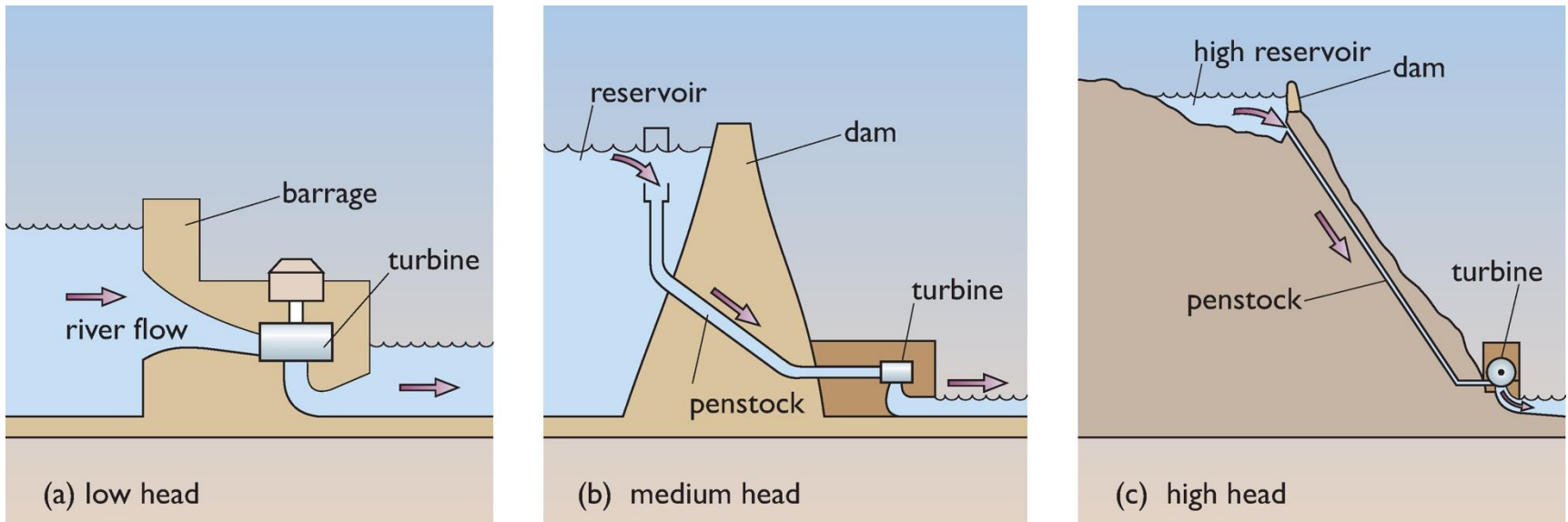
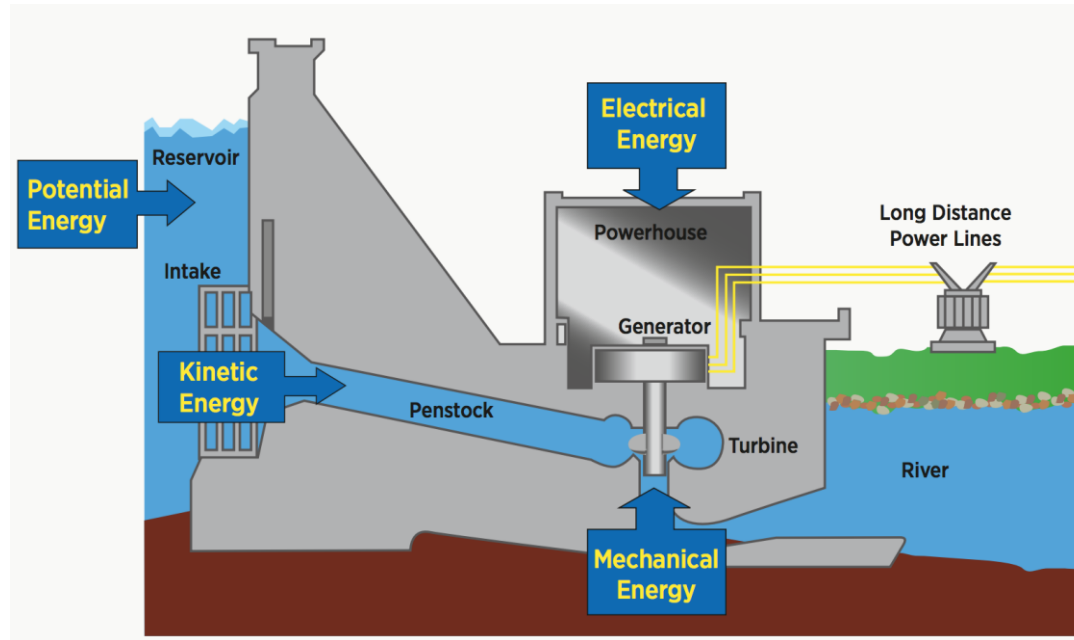


Fig. (1) low head ('run-of-river') and (2) high head often implies an effective head >100 meters

The higher head →

- the higher the water pressure across the hydro turbine and the more generated power;
- the higher water pressure means we can force a higher flow rate through a smaller turbine (often cost less).

Energy of PSH



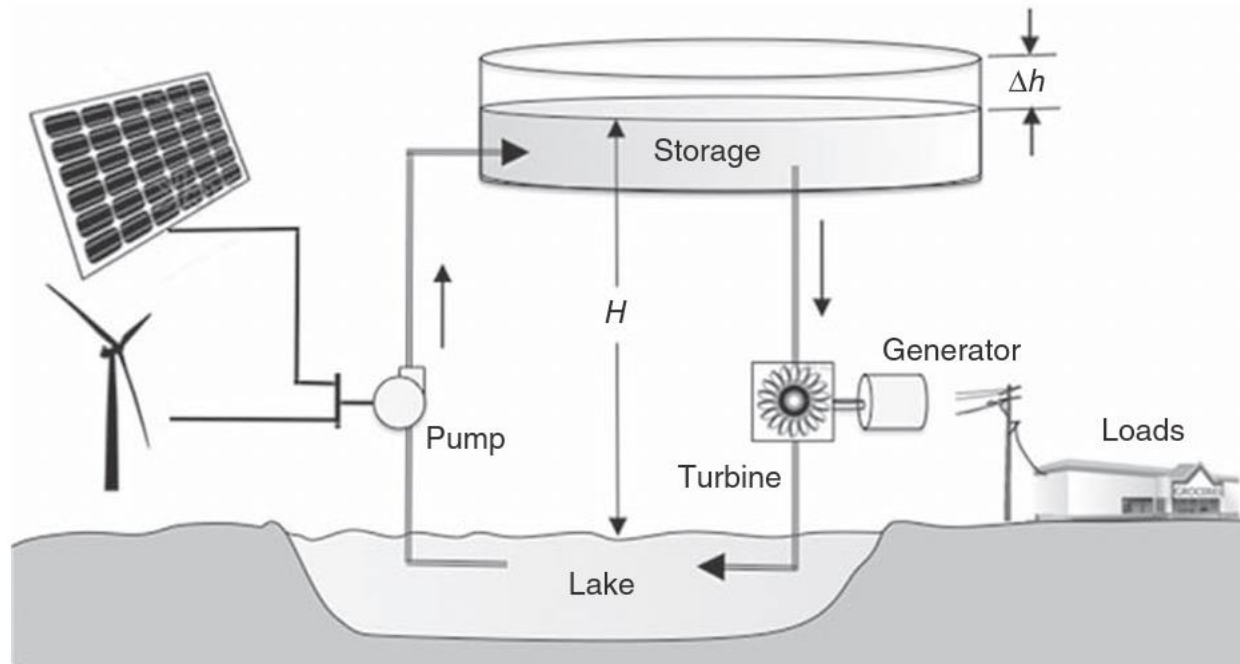
The energy available in the upper reservoir, relative to the lower one is given by

$$E = \frac{\rho A g \Delta h H}{3.6 \times 10^6}$$

where E is the energy (kWh), ρ is the density of water (1000 kg/m^3), A is the surface area of the upper reservoir (m^2), Δh is the allowable change in the surface level (m), g is gravitational acceleration (9.81 m/s^2), H is the average difference in elevation of the two reservoirs (m), and the 3.6 million converts J to kWh.

Example

Example 8.6 A Two-Penstock Pumped Storage System. The system in Figure 8.39 needs to supply a small village that has a 100-kW average load. A storage pond will be built at an elevation of 250 m above an existing lake. The pond will be 3-m deep but only the top 2 m can be used for variable storage. Assuming each penstock has an efficiency of 90% and the pump and turbine each have 85% efficiency, how big should the pond be to provide 2 days of storage during which time the renewables are unable to produce power?



Example Solution

Solution. Energy that needs to be delivered during those two days is

$$E = 100 \text{ kW} \times 24 \text{ h/d} \times 2 \text{ d} = 4800 \text{ kWh to the village}$$

Including inefficiencies, the tank needs to provide

$$E \text{ (from tank)} = \frac{4800 \text{ kWh}}{0.90 \times 0.85} = 6274.5 \text{ kWh}$$

$$E(kWh) = \frac{1000(kg/m^3)A(m^2)9.81(m/s^2)\Delta h(m)H(m)}{3.6 \times 10^6(J/kWh)}$$

$$A = \frac{3.6 \times 10^6 \times 6274.5}{1000 \times 9.81 \times 2 \times 250} = 4605.1 m^2$$

Flow Rate & Hydropower

$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \underbrace{\frac{\text{Weight}}{\text{Volume}}}_{\gamma} \times \underbrace{\frac{\text{Volume}}{\text{Time}}}_{Q} \times \underbrace{\frac{\text{Energy}}{\text{Weight}}}_{H}$$

$\gamma = \rho g$: **specific weight** (weight per unit volume)

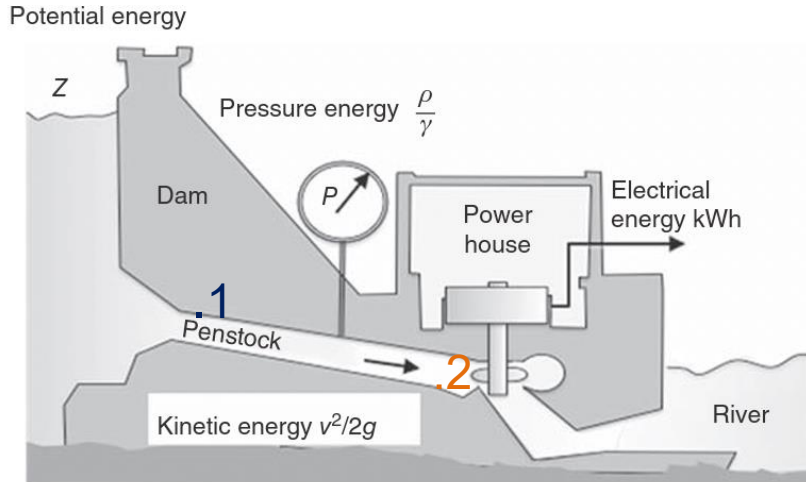
Q: **volume flow rate** (control the amount of water)

H: **net head** (control how hard the water pushes)

- Add turbine efficiency η

$$\begin{aligned} P_{\text{delivered}} \text{ (kW)} &= 9.81 \eta Q \text{ (m}^3/\text{s)} H_N \text{ (m)} \\ &= \frac{\eta Q \text{ (gpm)} H_N \text{ (ft)}}{5300} \end{aligned}$$

Fluid Dynamics: Bernoulli's Principle



$$z + \frac{p}{\gamma} + \frac{v^2}{2g} = \text{constant}$$

z = elevation above a reference height (m) or (ft)

p = pressure (N/m²) or (lb/ft²)

γ = specific weight (N/m³) or (lb/ft³)

v = average velocity (m/s) or (ft/s)

g = gravitational acceleration (9.81 m/s²) or (32.2 ft/s²)

potential head + pressure head + kinetic head = energy head

$$\text{energy head}_1 = \text{energy head}_2$$

Head (per-weight energy). The principle asserts that the sum of all forms of energy in a fluid is the same at all points on that streamline.

$$z = \frac{mgz}{mg} = \frac{\text{potential energy}}{\text{weight}}$$

$$\frac{p}{\gamma} = \frac{pV}{\rho Vg} = \frac{\text{pressure energy}}{\text{weight}}$$

$$\frac{v^2}{2g} = \frac{mv^2/2}{mg} = \frac{\text{kinetic energy}}{\text{weight}}$$

$$\left(p = \frac{F}{A} = \frac{F \times d}{A \times d} = \frac{\text{Energy}}{V} \right)$$

Example

TABLE 8.10 Useful Conversions for Water

	American	SI
1 cubic foot =	7.4805 gal	0.02832 m ³
1 foot per second =	0.6818 mph	0.3048 m/s
1 cubic foot per second =	448.8 gal/min(gpm)	0.02832 m ³ /s
Water density =	62.428 lb/ft ³	1000 kg/m ³
1 pound per square inch =	2.307 ft of water	6896 N/m ²
1 kW =	737.56 ft-lb/s	1000 N-m/s

Example: Suppose a 4-inch-diameter penstock delivers 150gal/min of water through an elevation change of 100 feet. The pressure in the pipe is 27 psi (pounds per square inch) when it reaches the powerhouse. What fraction of the available head is lost in the pipe? How much power is available for the turbine?

Solution. From Equation 8.16:

$$\text{Pressure head} = \frac{p}{\gamma} = \frac{27 \text{ lb/in}^2 \times 144 \text{ in}^2/\text{ft}^2}{62.428 \text{ lb/ft}^3} = 62.28 \text{ ft}$$

To find velocity head, we need to use $Q = vA$, where Q is flow rate, v is velocity, and A is cross-sectional area:

$$v = \frac{Q}{A} = \frac{150 \text{ gpm}}{(\pi/4) \cdot (4/12 \text{ ft})^2 \times 60 \text{ s/min} \times 7.4805 \text{ gal/ft}^3} = 3.83 \text{ ft/s}$$

So, from Equation 8.16, the velocity head is

$$\text{Velocity head} = \frac{v^2}{2g} = \frac{(3.83 \text{ ft/s})^2}{2 \times 32.2 \text{ ft/s}^2} = 0.228 \text{ ft}$$

Example Solution (cont'd)

The total head available for the turbine is the sum of the velocity and pressure head, or $62.28 + 0.228 \text{ ft} = 62.51 \text{ ft}$. This is called the net head, H_N .

$$H_N = 62.28 + 0.228 \text{ ft} = 62.51 \text{ ft of head}$$

Note that the velocity head is negligible.

Since we started with 100 ft of head, pipe losses are $100 - 62.51 \text{ ft} = 37.49 \text{ ft}$, or 37.49%.

Using conversions from Table 8.10, while carefully checking to see that units properly cancel, the power delivered to the turbine by a flow of 150 gal/min at a head of 62.51 ft is

$$P = \frac{150 \text{ gpm} \times 62.428 \text{ lb/ft}^3 \times 62.51 \text{ ft}}{60 \text{ s/min} \times 7.4805 \text{ gal/ft}^3} \times \frac{\text{kW}}{737.56 \text{ ft-lb/s}} = 1.77 \text{ kW}$$

Friction Head Loss

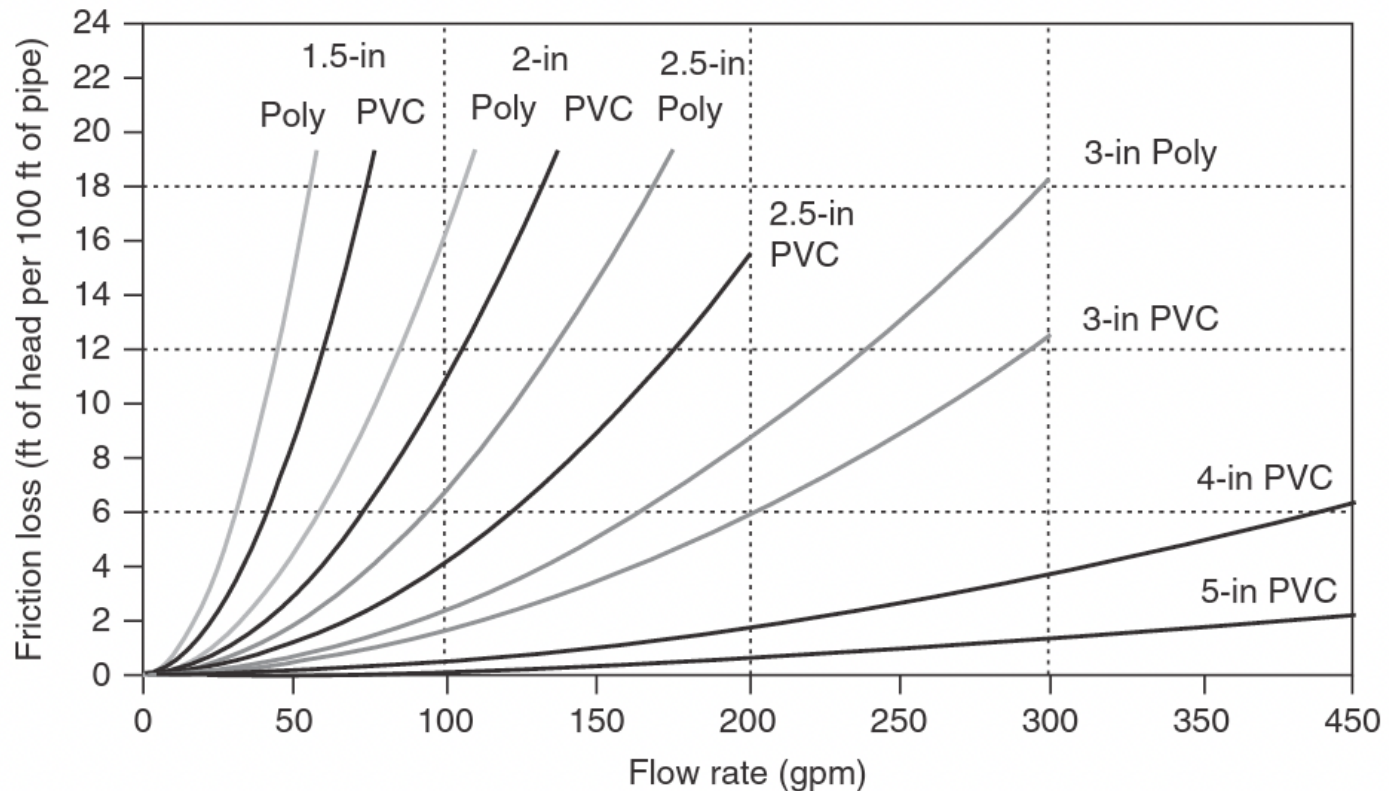


FIGURE 8.35 Friction head loss, in feet of head per 100 ft of pipe, for 160-psi PVC piping and for polyethylene, SDR pressure-rated pipe.

- Poly (polythene): small-diameter ones are easy to install.
- PVC (polyvinyl chloride): has lower friction losses and less expensive.
- Both need to be protected from sunlight that makes them brittle.

Example

Example 8.4 Power from a Micro-Hydro Plant. Suppose 150 gal/min of water is taken from a creek and delivered through 1000 feet of 3-in-diameter polyethylene pipe to a turbine 100 ft lower than the source. Assuming 80% efficiency for the turbine/generator, estimate the energy delivered in a 30-day month.

Solution. From Figure 8.35, at 150 gal/min, 3-in poly loses about 5 ft of head for every 100 ft of length. Since we have 1000 ft of pipe, the friction loss is

$$1000 \text{ ft} \times 5 \text{ ft}/100 \text{ ft} = 50 \text{ ft of head loss}$$

The net head available to the turbine is

$$\text{Net head} = \text{Gross head} - \text{Friction head} = 100 \text{ ft} - 50 \text{ ft} = 50 \text{ ft}$$

Using a 70% turbine/generator efficiency in Equation 8.20 suggests delivered power would be

$$P_{\text{delivered}} (\text{kW}) = \frac{\eta Q (\text{gpm}) H_N (\text{ft})}{5300} = \frac{0.80 \times 150 \times 50}{5300} = 1.13 \text{ kW}$$

The monthly electricity supplied would be $24 \text{ h} \times 30 \text{ d/mo} \times 1.13 \text{ kW} = 815 \text{ kWh}$, which is roughly the average electrical energy used by a typical U.S. household.

Hydropower Turbines



Turbines vary considerably in size, with 'runner' diameters ranging from as little as a 1/3 meters to some 20 times this.

We'll look at how they work, the factors that determine their efficiency, and the site parameters that determine the most suitable turbine.



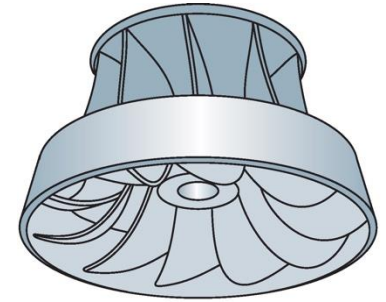
Impulse vs Reaction Turbines

Energy in water manifests itself in three forms – potential, pressure, and kinetic head. There are different approaches to transforming that waterpower into the mechanical energy needed to rotate the shaft of an electrical generator.

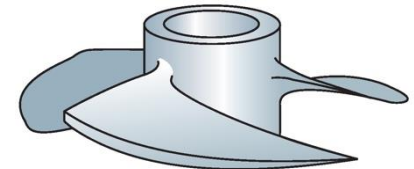
- **Impulse turbines:** capture the kinetic energy of high-speed jets of water squirting onto buckets along the circumference of a wheel.
- **Reaction turbines:** water velocity plays only a modest role, and instead it is mostly the pressure difference across the runners/ blades that creates the desired torque.
- *Impulse turbines are most appropriate in high-head, low-flow circumstances, while the opposite is the case for reaction turbines.*

Types of Turbine

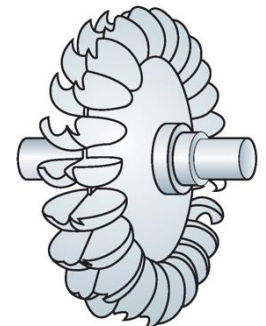
- **Francis turbines** – a combination of both impulse and reaction turbines, where the blades rotate using both reaction and impulse force of water: the most common type in medium/large-scale plants.
 - 1) they are radial-flow turbines → the water flow is *inwards* towards the center.
 - 2) maintain exactly the right speed and direction of the incoming water relative to the runner blades is important (95% efficiency).
- **‘Propeller’ or axial-flow turbines** sweep their blades through the entire area which the water enters → Good for very large volume flows and have become usual where the head is only a few meters.
- **Pelton wheels** (*impulse turbine*): operating in air at normal atmospheric pressure, and is a wheel with a set of double cups mounted around the rim.



Francis



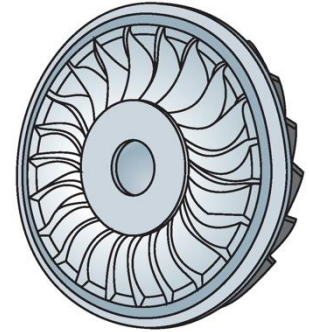
Fixed pitch propeller



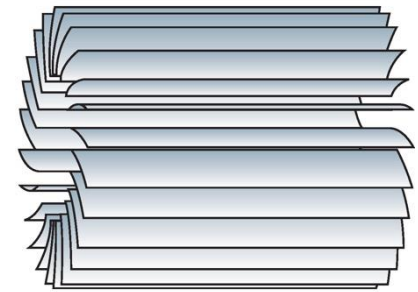
Pelton

Types of Turbine (Cont'd)

- **Turgo turbines** are a variant on the Pelton wheel, where the double cups are replaced by single, shallower ones. An impulse turbine can handle a larger volume of water than a Pelton wheel of the same diameter; an advantage for power generation at medium heads.
- **Cross-flow turbine** is another impulse type. The water enters as a flat sheet rather than a round jet. It is guided on to the blades, travels across the turbine and meets the blades a second time as it leaves.

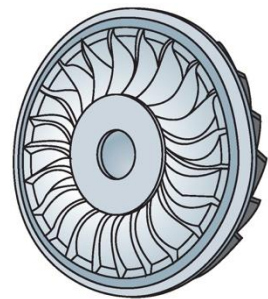
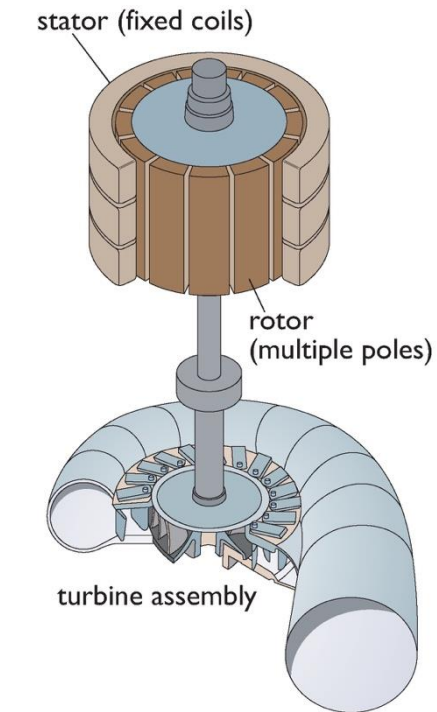


Turgo

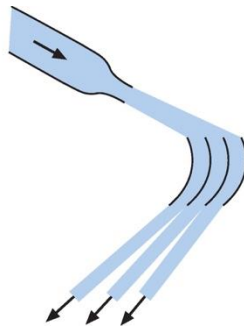


Crossflow

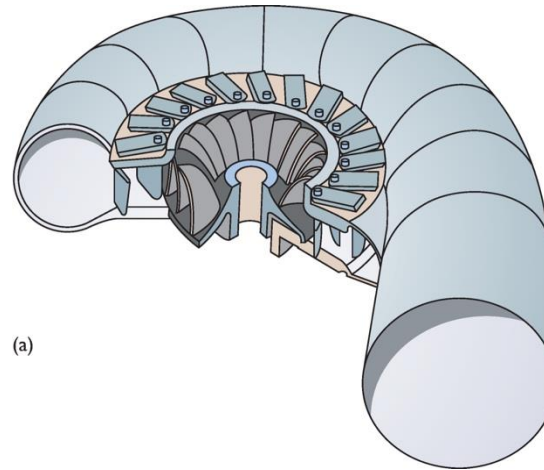
Turbine with Water Flows



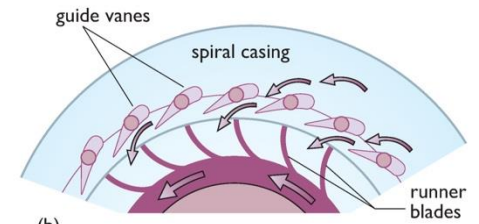
(a) runner



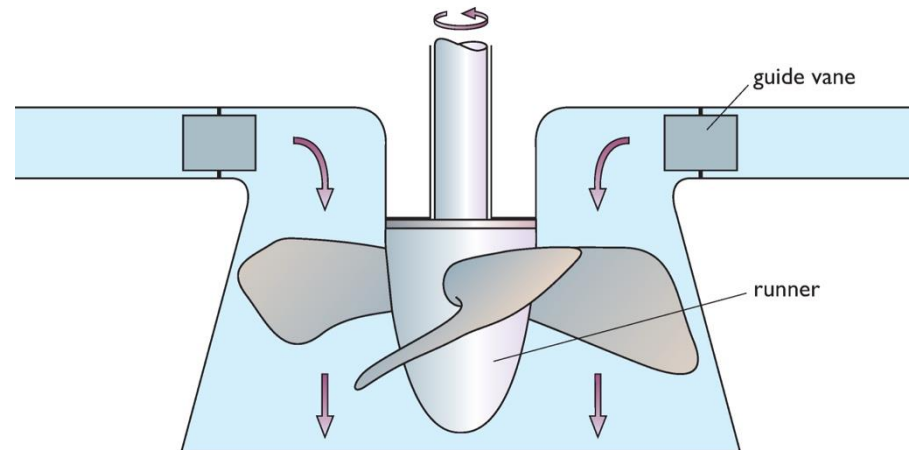
(b) water flow



(a)

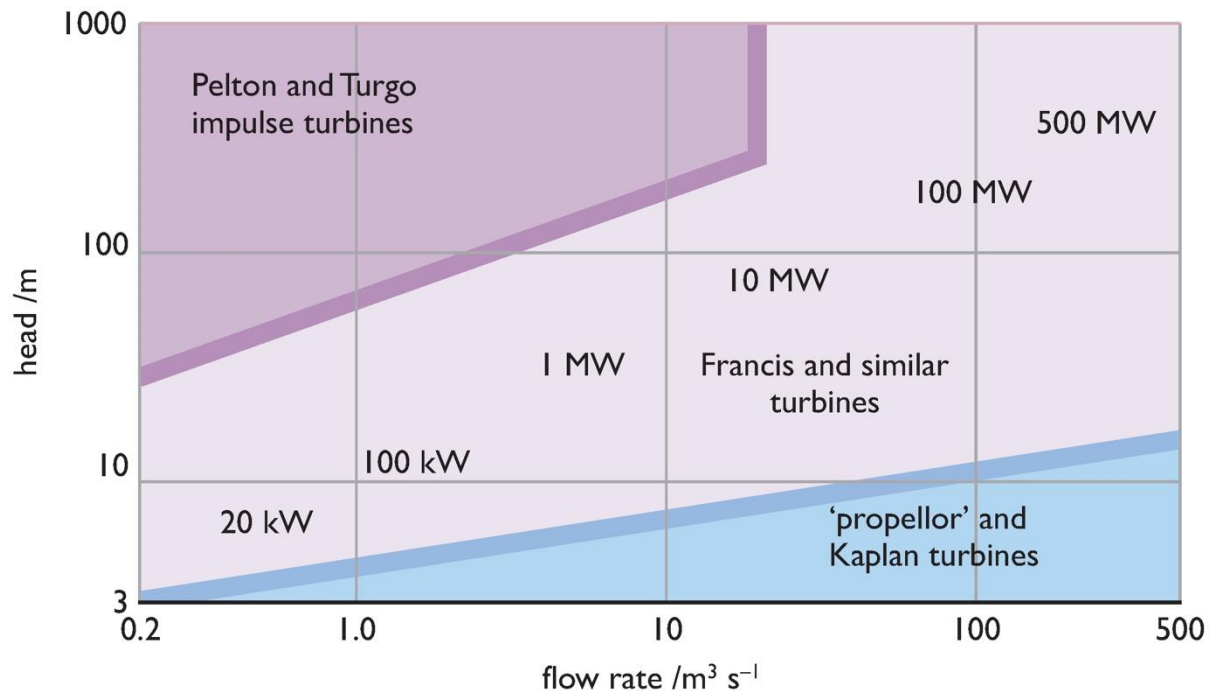


(b)



Ranges of Applications

Turbine	Head	Flow	Turbine Type	Efficiency	Typical Applications
Pelton	> 300 m	Low	Impulse	85–92%	Mountain hydro, high-pressure systems
Francis	30–300 m	Medium	Reaction	90–95%	Most medium/large hydro plants
Kaplan	2–30 m	High	Reaction	90–95%	Run-of-river, low-head dams



Pelton Turbine Design

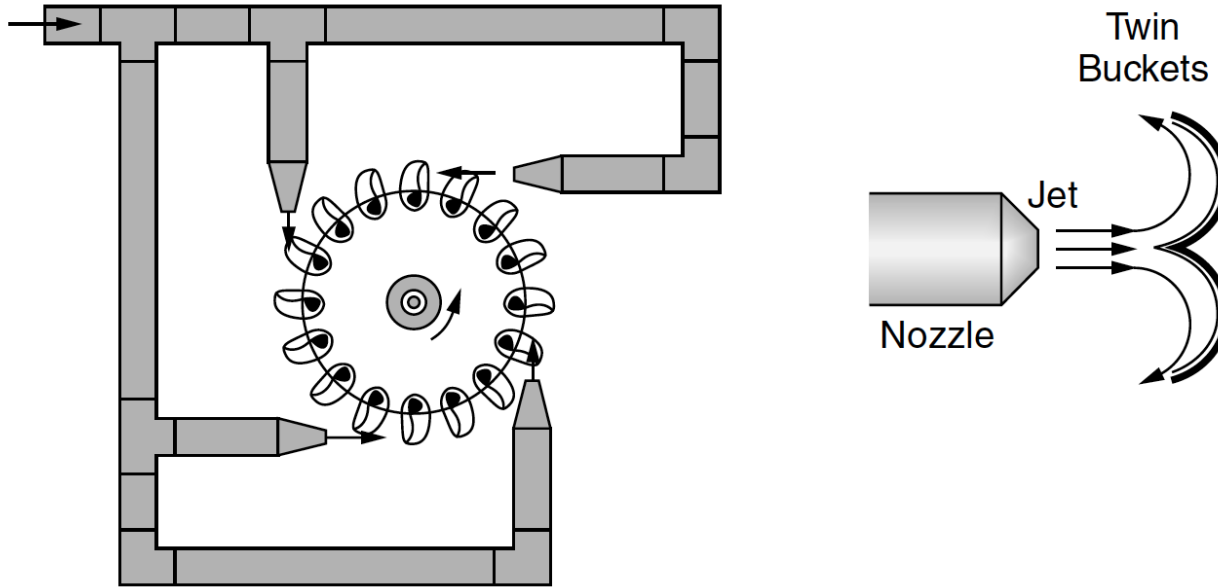


Figure: A four-nozzle Pelton turbine.

- Water squirts out of nozzles onto sets of twin buckets.
- The buckets are carefully designed to extract as much of the water's kinetic energy as possible while leaving enough energy in the water to enable it to leave the buckets w/o interfering with the incoming water.
- Typical efficiency: 70–90%.

Determine Jet Size

- Determine the flow velocity by the net head H_N :

$$H_N = \frac{v^2}{2g} \quad \text{so that} \quad v = \sqrt{2gH_N}$$

- Relate jet diameter with the flow rate Q : $Q = vA = \sqrt{2gH_N} \left(\frac{\pi}{4} \right) nd^2$

- Solving for jet diameter:

$$d = \frac{0.949}{(gH_N)^{1/4}} \sqrt{\frac{Q}{n}}$$

Q: A penstock provides 150 gpm (0.334 cfs) with 50 ft of head to a Pelton turbine with 4 nozzles. Assuming jet and nozzle diameters are the same, pick a nozzle diameter.

$$d = \frac{0.949}{(32.2 \text{ ft/s}^2 \times 50 \text{ ft})^{1/4}} \sqrt{\frac{0.334 \text{ ft}^3/\text{s}}{4}} = 0.0433 \text{ ft} = 0.52 \text{ in.}$$

Economics of Hydroelectricity

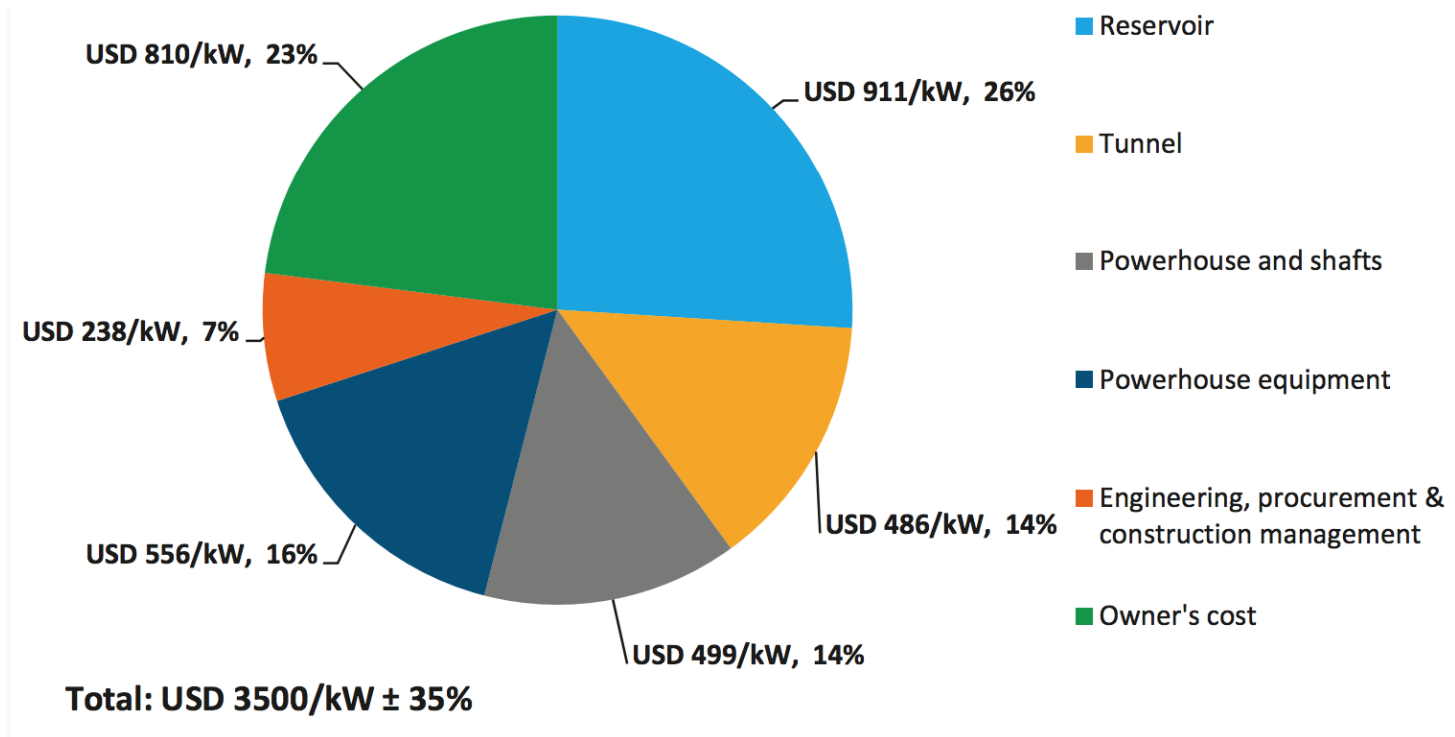


Fig: Cost breakdown of an indicative 500 MW Greenfield hydropower project.

- Hydroelectricity is well-established: the water-control systems, turbo-generators and output controls
- Expected lifetime of 20–25 years for the machinery, and 50–100 years for the external structures.

LCOE of Hydropower

Table: Typical installed costs and LCOE of hydropower projects

	Installed costs (USD/kW)	Operations and maintenance costs (%/year of installed costs)	Capacity factor (%)	Levelised cost of electricity (2010 USD/kWh)
Large hydro	1 050 – 7 650	2 – 2.5	25 to 90	0.02 – 0.19
Small hydro	1 300 – 8 000	1 – 4	20 to 95	0.02 – 0.27
Refurbishment/upgrade	500 – 1 000	1 – 6		0.01 – 0.05

Note: The levelised cost of electricity calculations assume a 10% cost of capital

Investment cost (USD/kW)	Discount rate (%)	LCOE (US cents/kWh)	Lifetime (years)	LCOE (US cents/kWh)
1 000	3	1.7	80	1.5
1 000	7	2.5	80	2.4
1 000	10	3.2	80	3.2
2 000	3	3.5	80	2.9
2 000	7	5.1	80	4.8
2 000	10	6.5	80	6.3
3 000	3	5.2	80	4.4
3 000	7	7.6	80	7.3
3 000	10	9.7	80	9.5

Note: base case assumes an economic life of 40 years, a 45% capacity factor and 2.5% of capital costs per year for O&M.

Source: IPCC, 2011.

Hydropower as Power Supply

Few large power stations operate in isolation, and the extent to which a proposed plant can form a useful part of a supply system is important with the ideal characteristics being:

- constant availability
- a reserve energy store to buffer variations in input
- no correlation in input variations between power stations
- rapid response to changing demand
- an input which matches annual variation in demand
- no sudden or unpredictable changes in input
- a location which does not require long transmission lines.

Environmental Impact

- In operation it releases no CO₂, and negligible quantities of the oxides of sulfur and nitrogen that lead to acid rain
- It produces no particulates or chemical compounds such as dioxins that are directly harmful to human health
- It emits no radioactivity
- Dams may collapse, but they will not cause major fires
- Associated with positive environmental effects such as flood control or irrigation, and in some cases, its development leads to a valued amenity.

Physical Effects

- **Hydrological:** The construction process of a major hydroelectric dam itself can cause widespread disturbance, and the effect on a fragile eco-system can be long-lasting.
- **Dam failures:** DoE study found that only 2400 of the country's 80,000 dams had hydroelectricity plants. It is not surprising to find only 5-6 hydroelectric plants in the world list of major dam failures.
- **Silt:** Silt accumulation build-up reduces the volume of stored water and consequently the hydro potential of a site.
- **Fish:** France has many dams constructed during the early 20th century on rivers previously used by Atlantic fish, and as licenses became due for renewal in the 1990s, stringent requirements were introduced for the construction of fish ladders or similar passages.
- **Methane:** The vegetable matter that would normally decay in the air to produce CO₂ could decay anaerobically under water to produce methane (CH₄).

Social Effects

Aspect	Criticism	Defense
Displacement of Communities	Dams often force the relocation of large populations, leading to loss of homes, culture, and livelihoods.	Resettlement programs can offer better infrastructure, schools, and health care.
Cultural Heritage Loss	Flooded areas may include sacred sites, historical landmarks, and indigenous lands.	Governments may document, preserve, or relocate cultural artifacts before flooding.
Equity and Social Justice	Marginalized and indigenous groups disproportionately bear the costs.	Public involvement and benefit-sharing frameworks can help ensure equity.
Livelihood Disruption	Dams alter river ecosystems, affecting fishing, farming, and grazing activities.	Dams can create new jobs in construction, tourism, and reservoir-based industries.
Health Impacts	Stagnant reservoir waters may increase diseases like malaria and schistosomiasis.	Proper water management and sanitation plans can mitigate health risks.
Access to Electricity	Often benefits urban/industrial areas, not displaced rural communities.	Electrification improves regional development and quality of life in the long term.
Public Participation	Projects are sometimes implemented without proper consultation.	Modern dam projects increasingly require stakeholder engagement and environmental impact assessments.